

Observation: Words such as **service, excellent, good, friendly, and experience** are highlighted based on their frequency.

PROCEDURE

1. Create the customer age and satisfaction score vectors.
2. Use `hist()` to visualize the distribution of customer ages.
3. Use `table()` and `pie()` to represent satisfaction scores.
4. Divide customers into age groups using `cut()`.
5. Use `barplot()` to create the stacked bar chart.
6. Load the `wordcloud` and `RColorBrewer` packages.
7. Split the feedback into individual words and count their frequency.
8. Generate the word cloud using the word frequencies.

RESULT

Customer feedback data was successfully visualized using a **histogram, pie chart, stacked bar chart, and word cloud**.

KEY R FUNCTIONS USED

- `hist()` – Creates a histogram.
- `table()` – Counts frequencies.
- `pie()` – Creates a pie chart.
- `cut()` – Creates age groups.
- `barplot()` – Creates a bar chart.
- `strsplit()` – Splits text into words.
- `wordcloud()` – Creates a word cloud.
- `brewer.pal()` – Provides color palettes.

SET 3 — EMPLOYEE PERFORMANCE EVALUATION

Dataset

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92

3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

PROGRAM 1 — Line Chart

Dataset

```
employee_id <- c(1, 2, 3, 4, 5)
```

```
department <- c("Sales", "HR", "Marketing", "Sales", "HR")
```

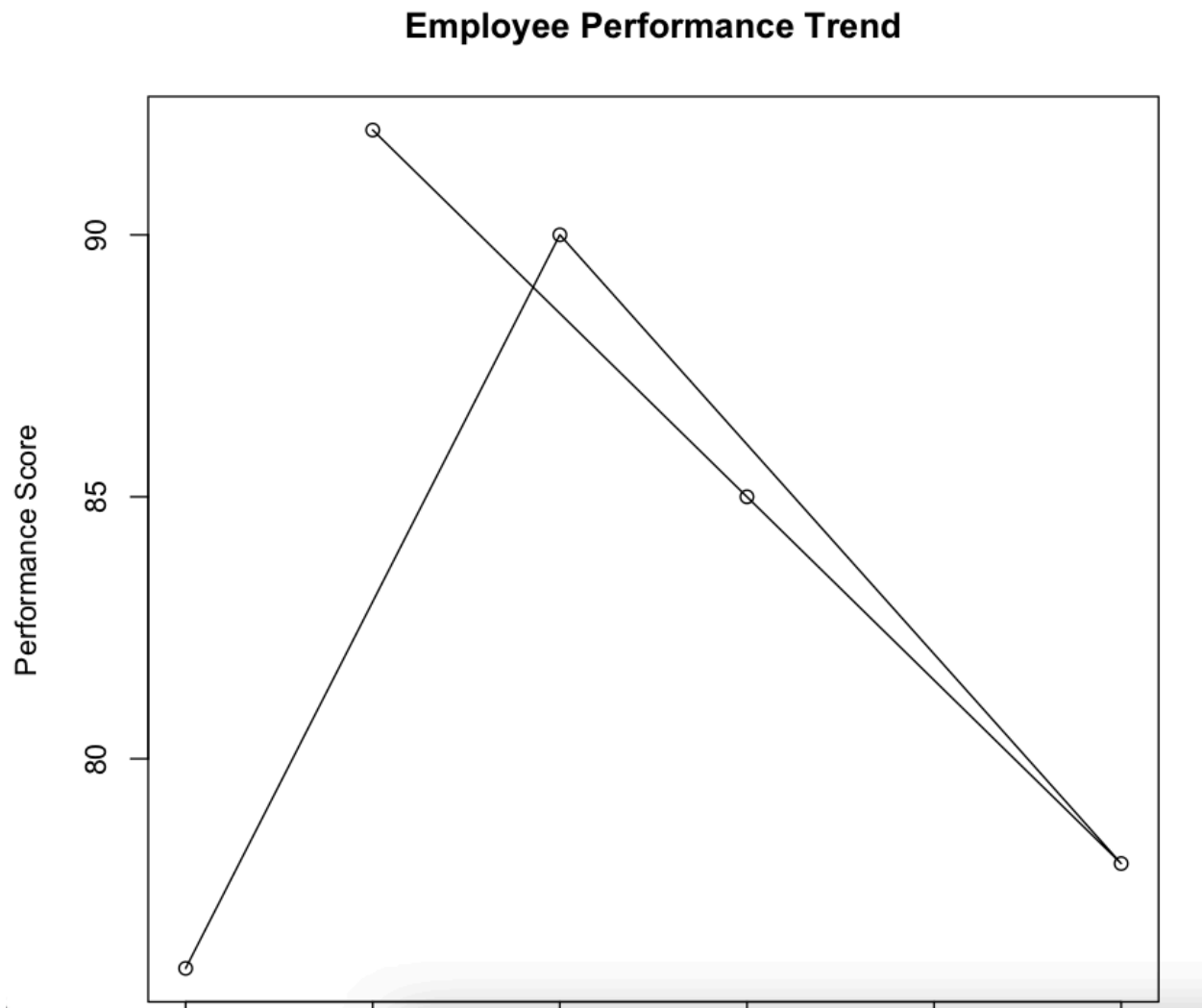
```
years <- c(5, 3, 7, 4, 2)
```

```
performance <- c(85, 92, 78, 90, 76)
```

Line Chart

```
plot(years, performance,
     type = "o",
     xlab = "Years of Service",
     ylab = "Performance Score",
     main = "Employee Performance Trend")
```

Output: Line chart showing employee performance according to years of service.

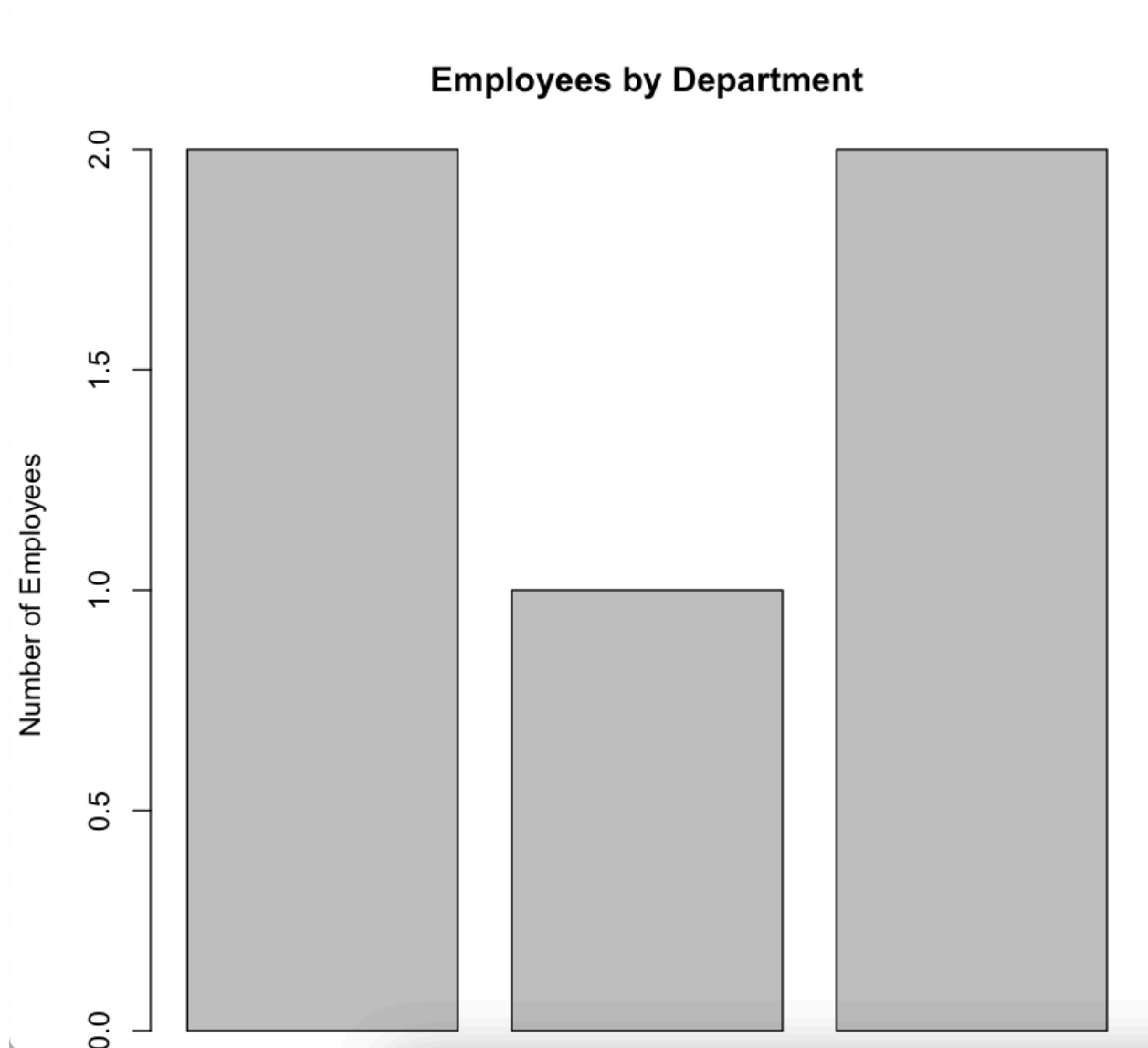


Observation: The chart shows the variation in performance scores with years of service.

PROGRAM 2 — Bar Chart

```
# Count employees in each department
department_count <- table(department)
# Bar Chart
barplot(department_count,
        xlab = "Department",
        ylab = "Number of Employees",
        main = "Employees by Department")
```

Output: Bar chart displaying the number of employees in each department.



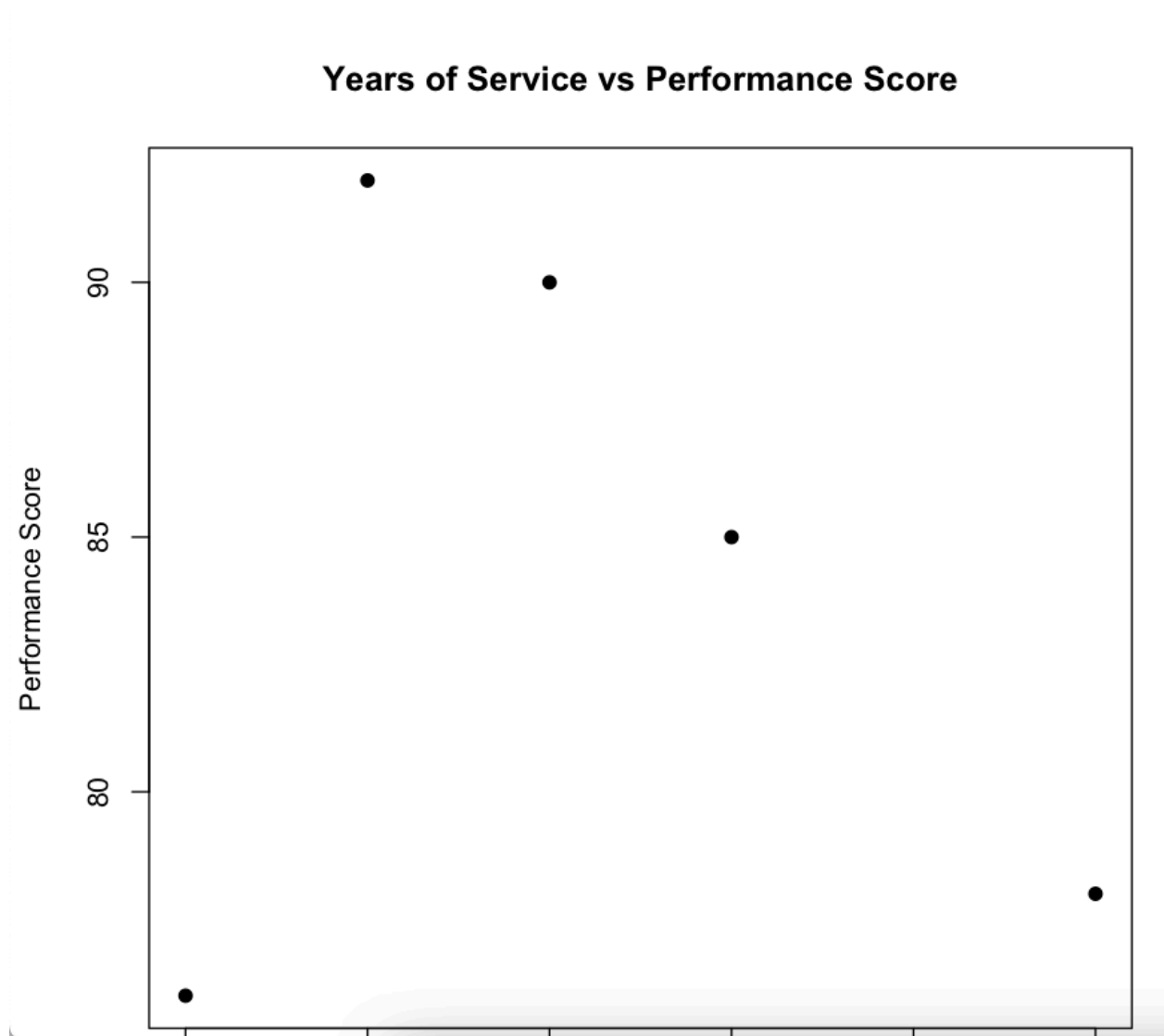
Observation: Sales and HR have two employees each, while Marketing has one employee.

PROGRAM 3 — Scatter Plot

```
# Scatter Plot
```

```
plot(years, performance,  
     pch = 19,  
     xlab = "Years of Service",  
     ylab = "Performance Score",  
     main = "Years of Service vs Performance Score")
```

Output: Scatter plot showing the relationship between years of service and performance score.



Observation: The scatter plot shows the distribution of employee performance scores with respect to years of service.

PROGRAM 4 — Interactive Dashboard

```
library(shiny)
```

```
# Dataset
```

```
employee_id <- c(1, 2, 3, 4, 5)
```

```
department <- c("Sales", "HR", "Marketing", "Sales", "HR")
```

```
years <- c(5, 3, 7, 4, 2)
```

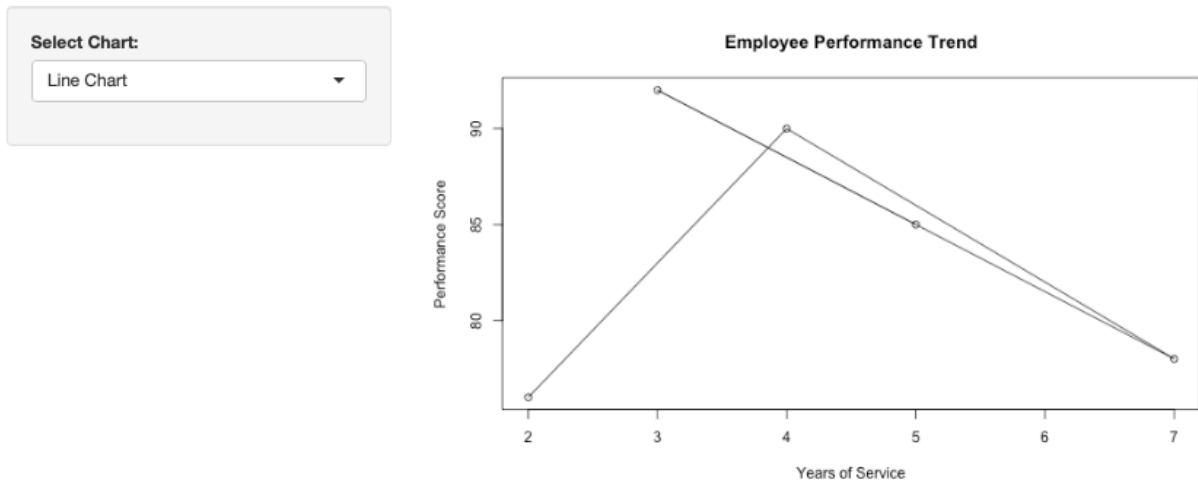
```

performance <- c(85, 92, 78, 90, 76)
department_count <- table(department)
# User Interface
ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput(
        "chart",
        "Select Chart:",
        choices = c("Line Chart", "Bar Chart")
      )
    ),
    mainPanel(
      plotOutput("employeePlot")
    )
  )
)
# Server
server <- function(input, output) {
  output$employeePlot <- renderPlot({
    if (input$chart == "Line Chart") {
      plot(years, performance,
           type = "o",
           xlab = "Years of Service",
           ylab = "Performance Score",
           main = "Employee Performance Trend")
    } else {
      barplot(department_count,
              xlab = "Department",
              ylab = "Number of Employees",
              main = "Employees by Department")
    }
  })
}
# Run Application
shinyApp(ui = ui, server = server)

```

Output: An interactive dashboard with a chart selection option. The user can select either **Line Chart** or **Bar Chart**.

Employee Performance Dashboard



Observation: The dashboard allows employee performance information to be viewed interactively using different charts.

PROCEDURE

1. Create vectors for employee ID, department, years of service, and performance score.
2. Use `plot()` to create the employee performance line chart.
3. Use `table()` and `barplot()` to display employees by department.
4. Use a scatter plot to show years of service versus performance score.
5. Create a Shiny interface using `fluidPage()` and `selectInput()`.
6. Use `renderPlot()` to display the selected chart.

7. Run the dashboard using shinyApp().

RESULT

Employee performance data was successfully visualized using **line chart, bar chart, and scatter plot**. An **interactive Shiny dashboard** was also created.

KEY R FUNCTIONS USED

- plot() – Creates line and scatter plots.
- table() – Counts employees by department.
- barplot() – Creates a bar chart.
- fluidPage() – Creates the Shiny interface.
- selectInput() – Provides chart selection.
- renderPlot() – Displays plots dynamically.
- shinyApp() – Runs the Shiny application.

SET 4 — PRODUCT INVENTORY MANAGEMENT

Dataset

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

PROGRAM 1 — Bar Chart

```
# SET 4: PRODUCT INVENTORY MANAGEMENT
# Dataset
product_id <- c(1, 2, 3, 4, 5)
product_name <- c(
  "Product A",
```